GKH Mohith

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Experience

Computepool Solutions **Hyderabad**
Full Stack Intern *Feb 2026 - present*

- Architecting and developing a full-stack food catering web application using HTML, CSS, and JavaScript for a responsive front-end.
- Designing the backend infrastructure using Node.js to handle user requests, menu data, and order processing.
- Collaborating within a small, agile team to build scalable web solutions and demonstrate core full-stack competencies.

Projects

FanOutService | *High-Performance Social Network Microservice (Ongoing) Link*

***Tech Stack**: C++, gRPC, Protocol Buffers, Redis, Docker, Multithreading*

- Architected a modular C++ backend engine replacing standard REST APIs with gRPC/Protobuf for high-speed, binary network communication.
- Implemented a Fanout-on-Write algorithmic feed architecture, utilizing a custom C++ std::thread pool for asynchronous background processing.
- Integrated Redis as the primary in-memory data store, using Hashes, Sets, and Lists to achieve $O(1)$ read latencies for user timelines.
- Containerized the application and database using Docker and docker-compose to ensure environment parity and fast local deployment.
- Successfully load-tested the gRPC server using ghz, achieving a stable throughput of 900 RPS with zero dropped connections under heavy concurrent load.

NodeLink | *P2P File Transfer Link*

***Tech Stack**: C++, CMake, Bencoding, BitTorrent, TCP and UDP sockets*

- Engineered a BitTorrent client parsing .torrent metadata and coordinating distributed downloads via TCP.
- Resolved critical runtime errors in legacy C++, including path resolution bugs and memory access violations.
- Reverse-engineered data flow from raw socket bytes to disk I/O, ensuring successful reassembly of ISO files.

DocAppoint App | *Serverless Appointment System Link Link*

***Tech Stack**: AWS (Lambda, API Gateway, Cognito, DynamoDB), NodeJS, React, Redux, CI/CD pipelines*

- Architected a scalable serverless app using AWS Lambda and React, reducing idle server costs to near zero.
- Engineered an RBAC system with AWS Cognito for secure data isolation across Patient, Doctor, and Admin.
- Automated deployment using Serverless Framework (IaC) for multi-environment provisioning.

Skills

Languages: C++, JavaScript, C

Tech Stack: React, redux, Next.js, Node.js, Express, MongoDB, SQL, cmake, Docker, redis, Git, AWS

Achievements

Paper on Hybrid Cloud Task Allocation accepted at ESPTA'24, IEEE MASS Conference, Seoul. Link

Education

SRMAP University
B.Tech (CSE), GPA: 8.39/10.0 *2021–2025*

Narayana Junior College
Class 12th (BIEAP), 94.2% *2019–2021*

Narayana Group of Schools
Class 10th (BSEAP), GPA: 10.0/10.0 *2018–2019*